

# AI Job Market Trend Analysis

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## Abstract

Artificial Intelligence (AI) is transforming industries by creating new job opportunities and increasing the demand for skilled professionals. Understanding job market trends is essential for organizations, job seekers, and researchers to make informed decisions. This project focuses on analyzing an AI Job Market Dataset containing 15,000 records using statistical analysis and data visualization techniques.

The analysis was performed through multiple phases including data understanding, descriptive statistical analysis, trend investigation, correlation analysis, hypothesis testing, and organizational segmentation. Various visualizations such as line charts, bar charts, scatter plots, and heatmaps were used to identify patterns and relationships among variables such as salary, experience level, company size, benefits score, and remote work opportunities.

The study provides valuable insights into AI hiring trends, salary distributions, organizational characteristics, and workforce requirements. The findings highlight the influence of experience and company size on compensation and reveal important patterns that can support recruitment strategies, workforce planning, and business decision-making. The project demonstrates how data analytics can be used to transform raw job market data into meaningful business intelligence.

# 1. Introduction

Artificial Intelligence (AI) has become one of the most influential technologies across industries. Organizations are increasingly adopting AI-driven solutions to improve operational efficiency, automate repetitive tasks, enhance decision-making processes, and gain competitive advantages. Understanding trends in AI adoption, salary patterns, company characteristics, and workforce requirements is essential for organizations, job seekers, and policymakers.

This project focuses on analyzing an AI Job Market Dataset containing 15,000 job records. Statistical analysis, visualization techniques, correlation analysis, hypothesis testing, and organizational segmentation were performed to identify patterns and generate meaningful business insights.

## 2. Problem Statement

The primary objective of this project is to analyze AI-related job market trends and understand the factors influencing salary, experience requirements, company characteristics, and employee benefits.

The project aims to:

- Understand the structure of the dataset.
- Perform descriptive statistical analysis.
- Investigate trends across industries and organizations.
- Analyze relationships between important variables.
- Conduct hypothesis testing.
- Segment organizations based on performance indicators.
- Generate actionable business recommendations.

### **3. Proposed Solution**

To address the problem of understanding AI job market trends, a data-driven analytical approach was implemented using Python and statistical techniques. The proposed solution involves collecting and analyzing job market data to identify meaningful patterns, trends, and relationships among various factors influencing employment opportunities in the AI sector.

The dataset was processed and examined through a structured analytical framework consisting of data exploration, statistical analysis, trend investigation, correlation analysis, hypothesis testing, and organizational segmentation. Visualization techniques were employed to represent trends clearly and support decision-making.

The proposed solution helps:

- Identify salary patterns across different experience levels and company sizes.
- Analyze industry-specific hiring trends.
- Understand the impact of benefits and organizational characteristics on compensation.
- Discover relationships between important job market variables.
- Generate actionable business insights and recommendations.

By applying statistical and visualization techniques, the solution enables organizations and stakeholders to make informed decisions regarding recruitment strategies, workforce planning, and talent management in the rapidly evolving AI industry.

### **4. Dataset Description**

The dataset contains 15,000 job postings related to Artificial Intelligence and Data Science roles.

## Features Used

- Job ID
- Job Title
- Salary (USD)
- Salary Currency
- Experience Level
- Employment Type
- Company Location
- Company Size
- Employee Residence
- Remote Ratio
- Required Skills
- Education Required
- Years of Experience
- Industry
- Posting Date
- Application Deadline
- Job Description Length
- Benefits Score
- Company Name

The dataset provides valuable information regarding organizational hiring practices, compensation structures, remote work opportunities, and workforce requirements.

## 5. Methodology

The project was completed through seven analytical phases:

### Phase 1: Data Understanding

- Examined dataset structure.
- Identified data types.
- Checked missing values.
- Reviewed feature distributions.

### Phase 2: Descriptive Statistical Analysis

Calculated:

- Mean
- Median
- Mode
- Standard Deviation
- Quartiles
- Percentiles

These measures helped summarize the characteristics of numerical variables

### Phase 3: Trend Investigation

The following trends were analyzed:

- Industry vs Salary
- Company Size vs Salary
- Experience Level vs Benefits Score

- Remote Work Ratio Analysis
- Hiring Pattern Trends

Visualization techniques included:

- Line Charts
- Bar Charts
- Scatter Plots
- Heatmaps

## **Phase 4: Correlation Analysis**

Relationships were examined between:

- Salary and Experience
- Benefits Score and Salary
- Company Size and Experience
- Remote Ratio and Benefits Score

Correlation matrices and heatmaps were used to understand variable dependencies.

## **Phase 5: Hypothesis Testing**

ANOVA statistical testing was performed.

### **Null Hypothesis ( $H_0$ ):**

Company size has no significant impact on salary distribution.

### **Alternative Hypothesis ( $H_1$ ):**

Company size significantly impacts salary distribution.

The test helped determine whether salary differences across company sizes were statistically significant.

## Phase 6: Organization Segmentation

Organizations were categorized into:

- High Salary Organizations
- Medium Salary Organizations
- Low Salary Organizations

Segmentation helped identify differences in compensation patterns and organizational behavior.

## Phase 7: Business Insights

**Factors affecting salary and job demand:** Experience level, company size, and industry.

**Factors increasing job attractiveness:** Higher benefits, advanced AI skills, and flexible work options.

**Impact of remote work:** Remote work opportunities improve job attractiveness.

**Less competitive organizations:** Organizations with lower salaries, limited benefits, and fewer remote work options.

### Recommendations:

- Offer competitive salaries.
- Improve employee benefits.
- Expand remote work opportunities.
- Key findings were converted into business recommendations for strategic decision-making.

## 6. Statistical Findings

The analysis revealed several important observations:

- Salary generally increases with experience level.
- Larger organizations tend to provide higher compensation packages.
- Benefits scores show a positive relationship with salary levels.
- Remote work opportunities vary across industries.
- Some industries demonstrate stronger demand for AI professionals.

Descriptive statistics showed noticeable variation in salary distribution, indicating differences among industries, job roles, and organization types.

## 7. Trend Analysis Results

### Industry Analysis

Certain industries showed higher demand for AI professionals and offered more competitive salaries.

### Company Size Analysis

Large organizations consistently demonstrated stronger compensation packages compared to smaller organizations.

### Experience Analysis

Professionals with greater years of experience were associated with significantly higher salary levels.

### Remote Work Analysis

Remote work opportunities differed among industries, highlighting changing workplace preferences and organizational policies.

## 8. Visual Insights

The visualizations provided clear evidence of:

- Positive salary growth with increasing experience.
- Salary variation among company sizes.
- Relationships between employee benefits and compensation.
- Industry-specific hiring trends.
- Workforce distribution across different organizational categories.

Graphs and heatmaps enabled effective interpretation of complex relationships within the dataset.

## 9. Business Recommendations

Based on the analysis, the following recommendations are proposed:

1. Increase investment in employee skill development programs.
2. Offer competitive salary packages to attract experienced professionals.
3. Enhance employee benefits to improve talent retention.
4. Expand remote work opportunities where feasible.
5. Focus recruitment efforts on high-growth AI sectors.
6. Benchmark compensation strategies against industry leaders.

## 10. Future Scope

Future enhancements may include:

- Machine Learning-based salary prediction.
- AI job demand forecasting.

- Skill-gap analysis.
- Country-wise job market comparison.
- Interactive dashboard development using Power BI or Tableau.
- Real-time labor market analytics

## 11. REFERENCES

[1] Kaggle, AI Job Market and Salary Trends Dataset.

[2] Pandas Documentation, <https://pandas.pydata.org>

[3] NumPy Documentation, <https://numpy.org>

[4] Matplotlib Documentation, <https://matplotlib.org>

[5] Seaborn Documentation, <https://seaborn.pydata.org>

# Appendix A – Dataset Overview

## CODE

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv(r"C:\Users\Poonguzhali\OneDrive\Attachments\Desktop\ai_job_dataset.csv")
```

```
df.head()
```

## OUTPUT

|   | job_id  | job_title                   | salary_usd | salary_currency | experience_level | employment_type | company_location |
|---|---------|-----------------------------|------------|-----------------|------------------|-----------------|------------------|
| 0 | AI00001 | AI<br>Research<br>Scientist | 90376      | USD             | SE               | CT              | China            |
| 1 | AI00002 | AI<br>Software<br>Engineer  | 61895      | USD             | EN               | CT              | Canada           |
| 2 | AI00003 | AI<br>Specialist            | 152626     | USD             | MI               | FL              | Switzerland      |
| 3 | AI00004 | NLP<br>Engineer             | 80215      | USD             | SE               | FL              | India            |
| 4 | AI00005 | AI<br>Consultant            | 54624      | EUR             | EN               | PT              | France           |

```

: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15000 entries, 0 to 14999
Data columns (total 19 columns):
 #   Column                                Non-Null Count  Dtype  
---  -
 0   job_id                               15000 non-null  object 
 1   job_title                             15000 non-null  object 
 2   salary_usd                           15000 non-null  int64  
 3   salary_currency                       15000 non-null  object 
 4   experience_level                       15000 non-null  object 
 5   employment_type                       15000 non-null  object 
 6   company_location                      15000 non-null  object 
 7   company_size                          15000 non-null  object 
 8   employee_residence                    15000 non-null  object 
 9   remote_ratio                          15000 non-null  int64  
10  required_skills                       15000 non-null  object 
11  education_required                    15000 non-null  object 
12  years_experience                       15000 non-null  int64  
13  industry                              15000 non-null  object 
14  posting_date                          15000 non-null  object 
15  application_deadline                  15000 non-null  object 
16  job_description_length                 15000 non-null  int64  
17  benefits_score                         15000 non-null  float64 
18  company_name                          15000 non-null  object 
dtypes: float64(1), int64(4), object(14)
memory usage: 2.2+ MB

```

**Figure A1:** Dataset overview showing sample records and feature information.

## Appendix B: Trend Analysis

### CODE

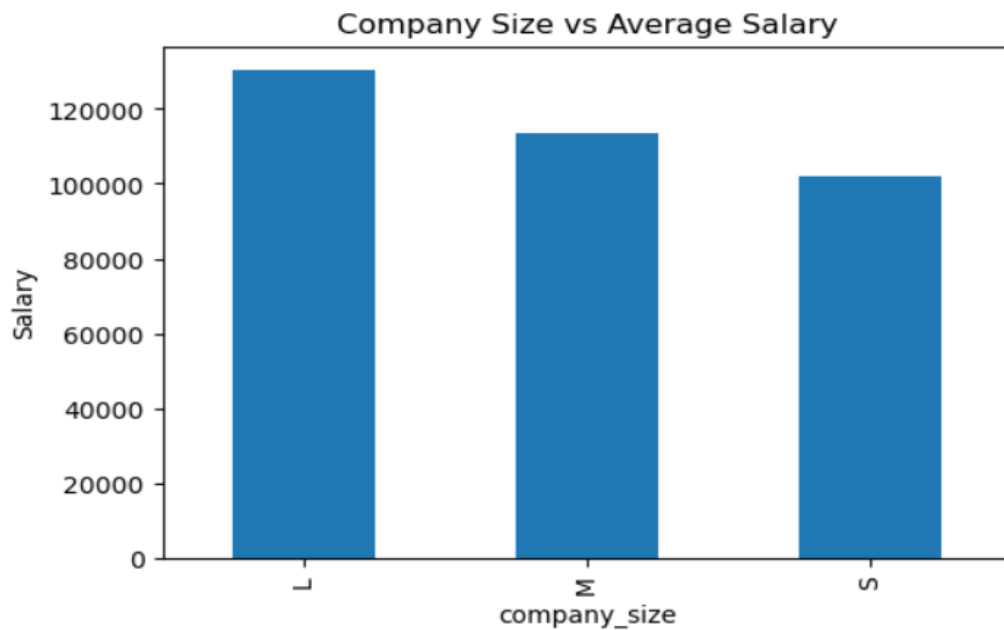
```

company_salary = df.groupby('company_size')['salary_usd'].mean()

plt.figure(figsize=(6,4))
company_salary.plot(kind='bar')
plt.title("Company Size vs Average Salary")
plt.ylabel("Salary")
plt.show()

```

## OUTPUT-BARCHART



**Figure B1:** Industry-wise salary trend analysis through Barchart.

## CODE

```
heat_data = pd.pivot_table(  
    df,  
    values='benefits_score',  
    index='experience_level',  
    columns='company_size',  
    aggfunc='mean'  
)  
  
plt.figure(figsize=(8,5))  
sns.heatmap(heat_data, annot=True)  
plt.title("Experience Level vs Benefits Score")  
plt.show()
```

## OUTPUT-HEATMAP



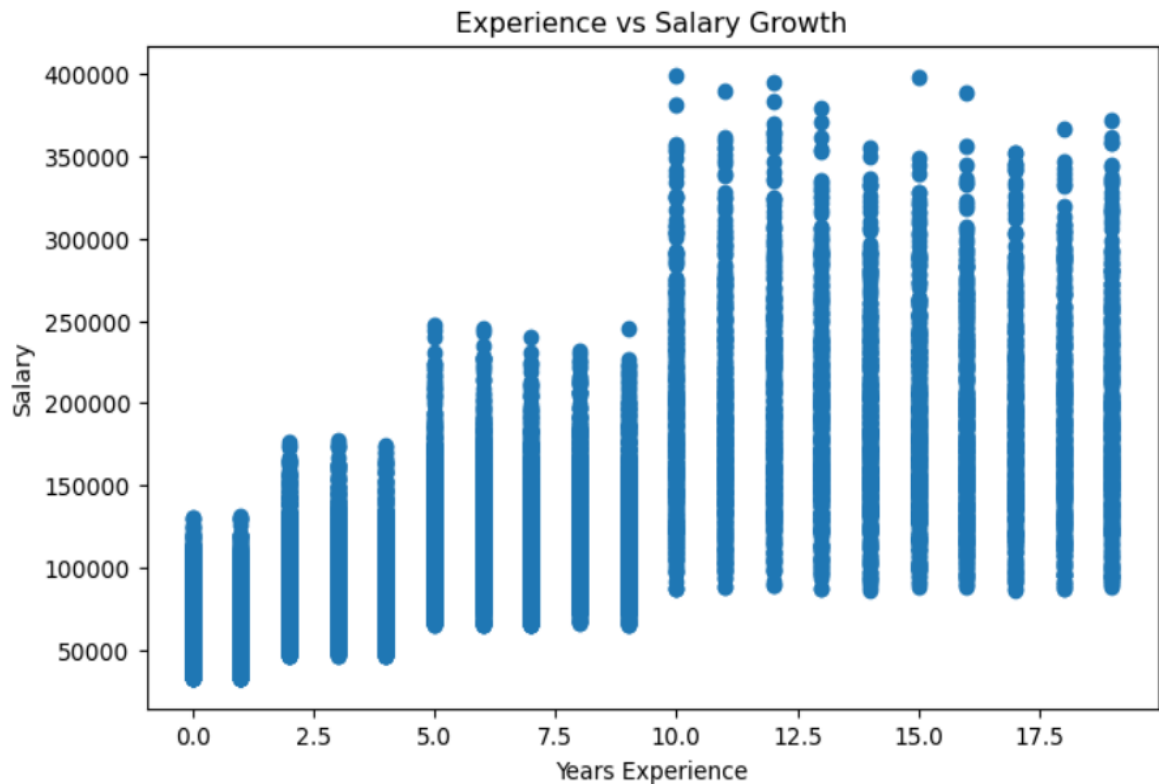
**Figure B2:** Industry-wise salary trend analysis through Heatmap.

## CODE

```
plt.figure(figsize=(8,5))
plt.scatter(df['years_experience'],
            df['salary_usd'])

plt.xlabel("Years Experience")
plt.ylabel("Salary")
plt.title("Experience vs Salary Growth")
plt.show()
```

## OUTPUT-SCATTERPLOT



**Figure B3:** Industry-wise salary trend analysis through Scatterplot.

## Appendix C: Correlation Analysis

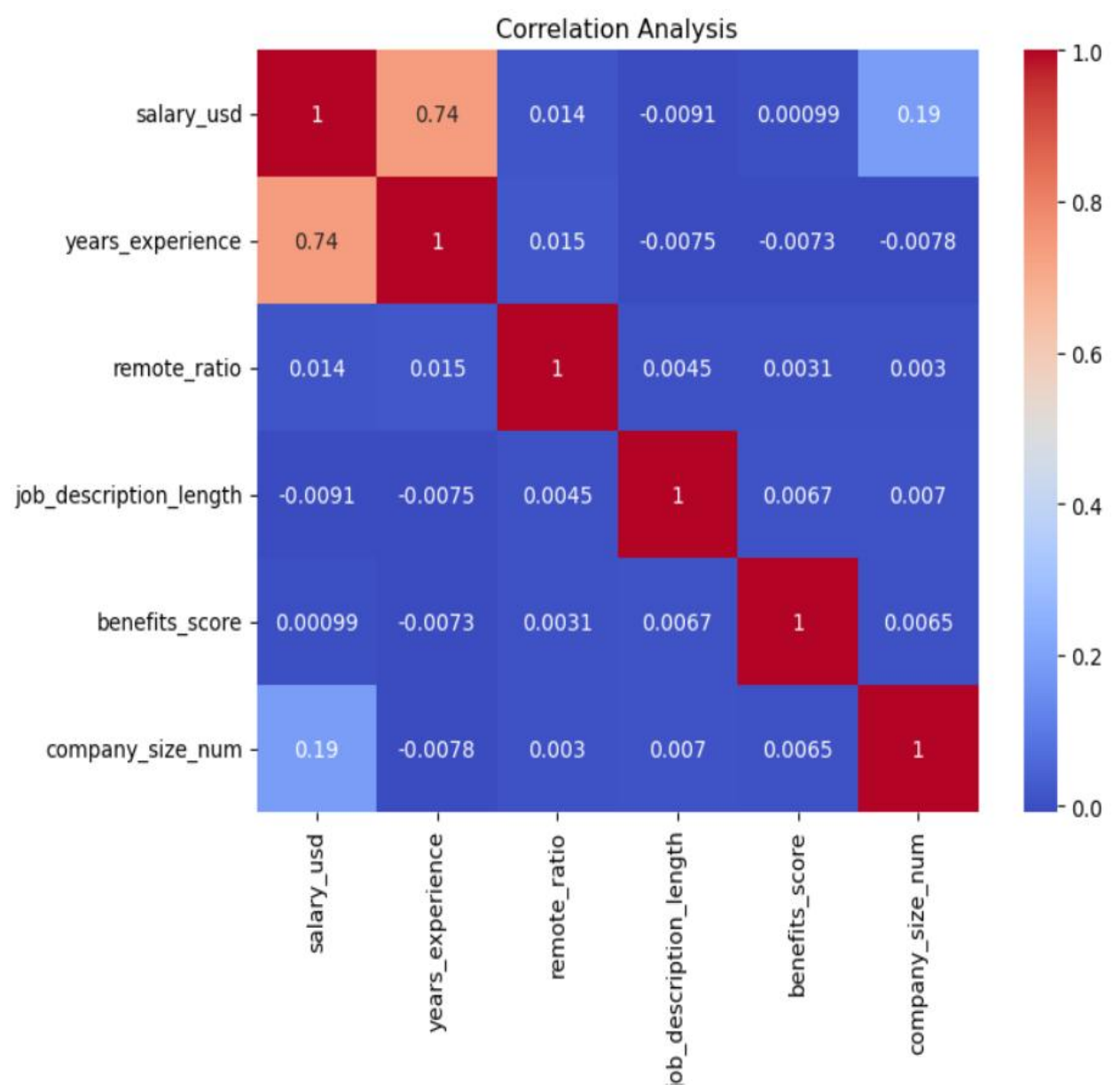
### CODE

```
plt.figure(figsize=(8,6))

sns.heatmap(corr,
            annot=True,
            cmap='coolwarm')

plt.title("Correlation Analysis")
plt.show()
```

# OUTPUT



**Figure C1:** Correlation matrix showing relationships among numerical variables.

## Appendix D: Hypothesis Testing (ANOVA)

### CODE

```
from scipy.stats import f_oneway

small = df[df['company_size']=='S']['salary_usd']
medium = df[df['company_size']=='M']['salary_usd']
large = df[df['company_size']=='L']['salary_usd']

f_stat, p_value = f_oneway(small, medium, large)

print("F-Statistic =", f_stat)
print("P-Value =", p_value)
if p_value < 0.05:
    print("Reject Null Hypothesis")
    print("Company size significantly impacts salary")
else:
    print("Fail to Reject Null Hypothesis")
    print("Company size does not significantly impact salary")
```

### OUTPUT

F-Statistic = 287.10387515940585

P-Value = 4.366865535018369e-123

Reject Null Hypothesis

Company size significantly impacts salary

## Appendix E: Organization Segmentation

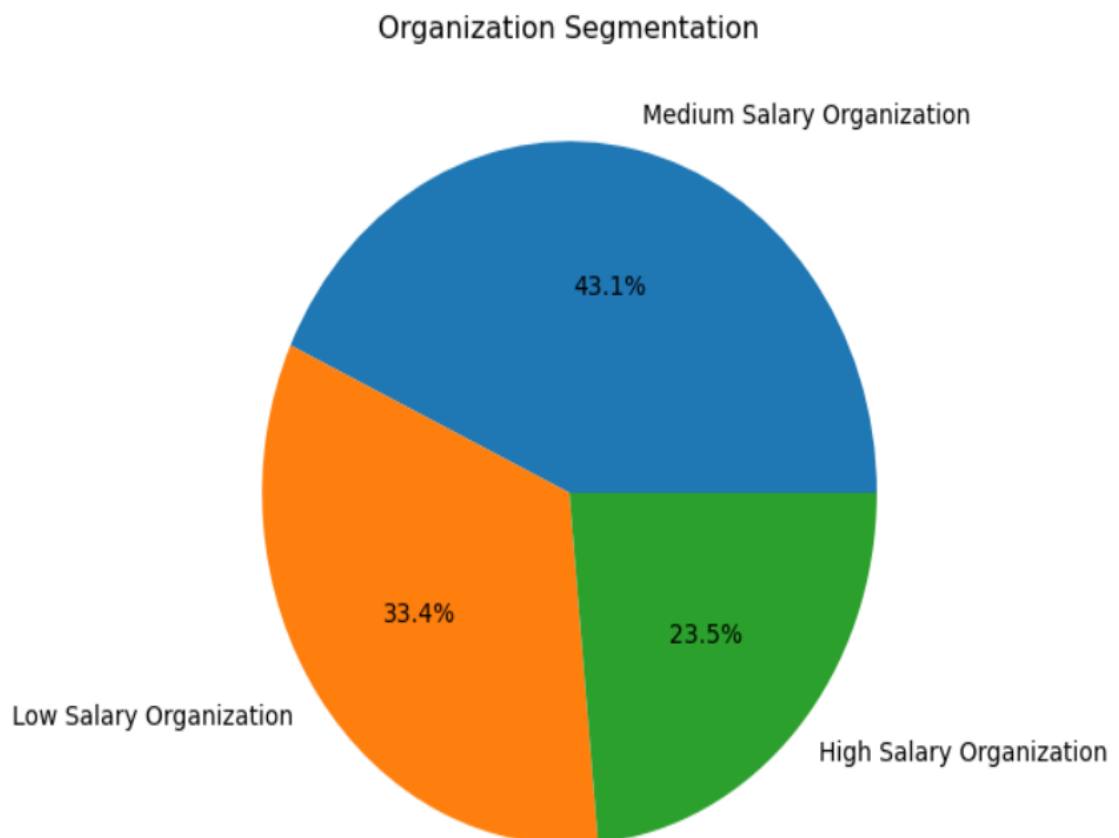
### CODE

```
: plt.figure(figsize=(6,6))

df['Organization_Category'].value_counts().plot(
    kind='pie',
    autopct='%1.1f%%'
)

plt.title("Organization Segmentation")
plt.ylabel("")
plt.show()
```

### OUTPUT



## 12.Conclusion

This project successfully analyzed AI job market trends using statistical analysis, visualization techniques, correlation analysis, hypothesis testing, and organizational segmentation. The findings demonstrate the influence of experience, company characteristics, and benefits on salary structures and hiring trends.

The generated insights can support organizations in workforce planning, recruitment strategy development, and compensation management. The project highlights the growing importance of AI talent and provides a foundation for future predictive analytics and decision-support systems.

## 13.Tools and Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- SciPy
- Jupyter Notebook
- GitHub

## GitHub Repository

AI-Trend-Analysis

<https://github.com/poonguzhalim/AI-Trend-Analysis.git>